

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

ENTITY RELATIONSHIP DIAGRAM

UPI Enablement Platform

Logical Data Model — 26 Entities Across 3 Functional Groupings

Document Version: 1.0 (Draft)

Classification: Internal / Confidential

Prepared by: Aniket Saraf

Date: July 2026

Table of Contents

Table of Contents 2

1. Introduction 3

 1.1 Notation Key 3

2. Core Profile & Account Entities 4

 2.1 Relationship & Cardinality Summary 4

3. Transaction & Payment Entities 6

 3.1 Relationship & Cardinality Summary 6

4. Support, Notification & Admin Entities..... 8

 4.1 Relationship & Cardinality Summary 8

1. Introduction

This document presents the logical Entity Relationship Diagram (ERD) for the Deutsche Bank India UPI Enablement Platform, elaborating the 26-entity Data Dictionary published in the BRD into three functionally grouped diagrams: Core Profile & Account, Transaction & Payments, and Support/Notification/Admin. Each entity is shown with its primary key (PK) and foreign key (FK) attributes; full attribute-level detail (data types, constraints, indexing) is maintained by the Database Team in the physical schema design, which this logical model informs.

1.1 Notation Key

Notation	Meaning
PK	Primary Key — uniquely identifies each record in the entity
FK	Foreign Key — references the primary key of a related entity
1	Exactly one related record (mandatory, single)
N	Zero or more related records (many)
0..1	Zero or one related record (optional, single)
1..N	One or more related records (mandatory, many)

Entities that participate in relationships across more than one diagram (e.g., UPI_Profile, Account, Transaction, Customer) are repeated with their key attributes in each relevant diagram for readability, rather than requiring a single unreadably dense diagram.

2. Core Profile & Account Entities

This diagram models the customer's identity, accounts, UPI registration profile, device bindings, PIN metadata, consent records, and saved beneficiaries — the foundational entities created during and after registration.

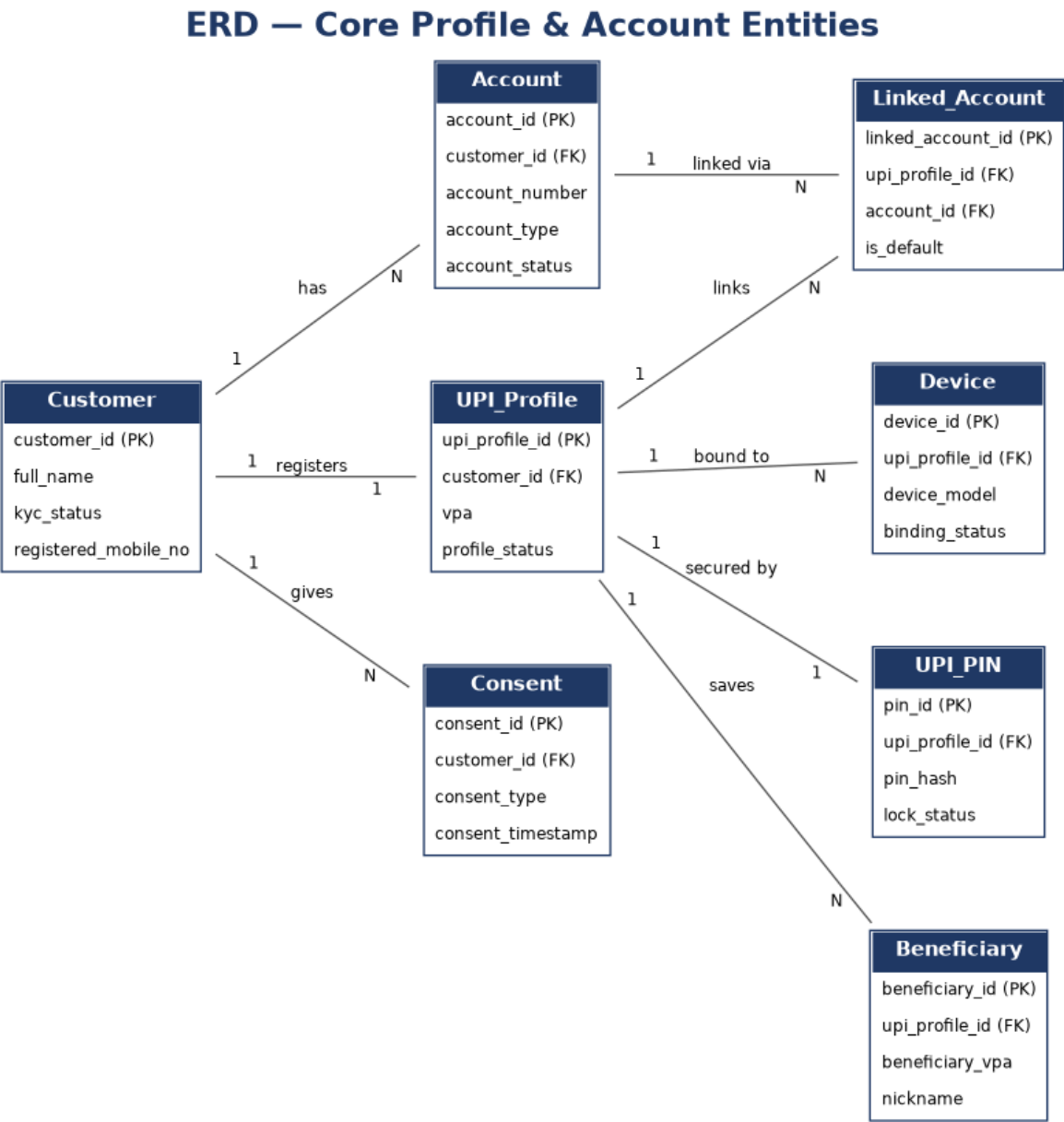


Figure 1 — ERD: Core Profile & Account Entities

2.1 Relationship & Cardinality Summary

Relationship	Cardinality	Business Meaning
Customer → Account	1 : N	A customer may hold multiple bank accounts
Customer → UPI_Profile	1 : 1	A customer has at most one active UPI profile
Customer → Consent	1 : N	A customer may give consent multiple times (e.g., re-consent on policy update)

Relationship	Cardinality	Business Meaning
UPI_Profile → Linked_Account	1 : N	A UPI profile links one or more eligible accounts
Account → Linked_Account	1 : N	An account may be linked to a UPI profile via one linking record
UPI_Profile → Device	1 : N	A profile may have historical device bindings (only one Active at a time)
UPI_Profile → UPI_PIN	1 : 1	A profile has exactly one current PIN record
UPI_Profile → Beneficiary	1 : N	A profile may save multiple favourite payees

3. Transaction & Payment Entities

This diagram models the core transaction record and its associated debit/credit legs, collect requests and reminders, QR codes and merchant validation, and the fraud/AML screening results attached to each transaction.

ERD — Transaction & Payment Entities

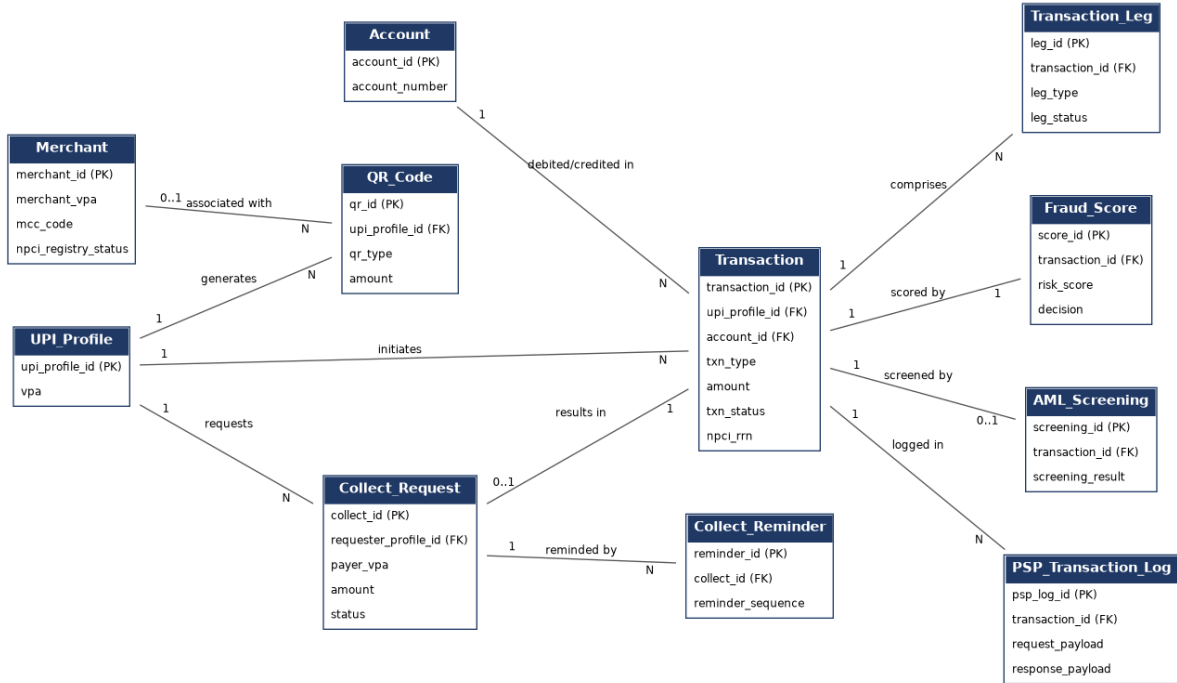


Figure 2 — ERD: Transaction & Payment Entities

3.1 Relationship & Cardinality Summary

Relationship	Cardinality	Business Meaning
UPI_Profile → Transaction	1 : N	A profile initiates many transactions over time
Account → Transaction	1 : N	An account is debited or credited across many transactions
Transaction → Transaction_Leg	1 : N	A transaction comprises one or more debit/credit legs for reconciliation
UPI_Profile → Collect_Request	1 : N	A profile may raise many collect requests as requester
Collect_Request → Collect_Reminder	1 : N	A request may have up to 3 reminders (BR-COL-04)
Collect_Request → Transaction	0..1 : 1	An approved collect request results in exactly one transaction
UPI_Profile → QR_Code	1 : N	A profile may generate multiple static/dynamic QR codes
Merchant → QR_Code	0..1 : N	A merchant QR code is associated with one merchant record

Relationship	Cardinality	Business Meaning
Transaction → Fraud_Score	1 : 1	Every transaction receives exactly one risk-scoring result
Transaction → AML_Screening	1 : 0..1	A transaction is screened for AML only where policy thresholds apply
Transaction → PSP_Transaction_Log	1 : N	A transaction's raw PSP request/response is logged for reconciliation

4. Support, Notification & Admin Entities

This diagram models customer notifications, the dispute lifecycle with investigation notes and refunds, the immutable audit log, business-configurable limits, and the internal RBAC model (roles, permissions, internal users).

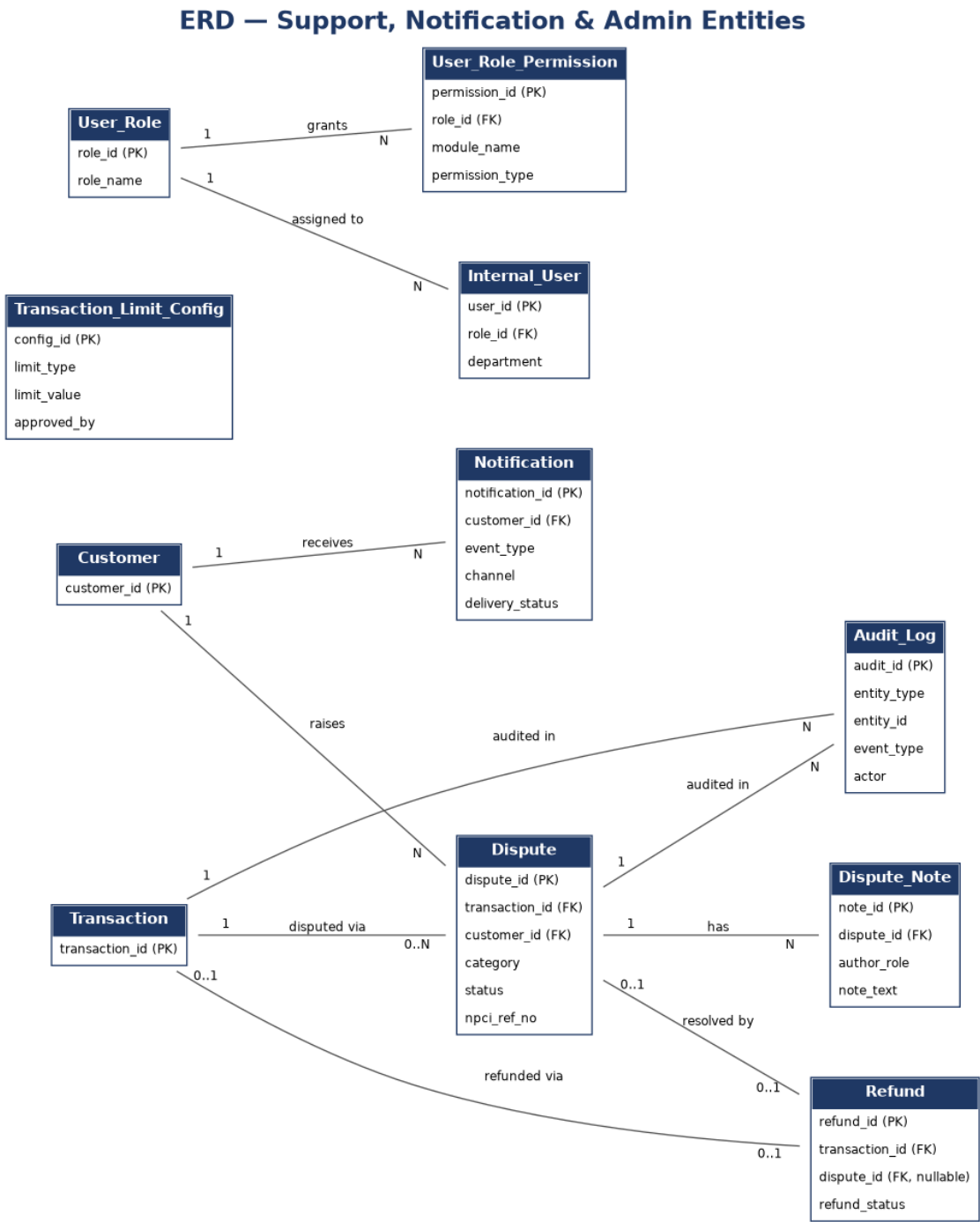


Figure 3 — ERD: Support, Notification & Admin Entities

4.1 Relationship & Cardinality Summary

Relationship	Cardinality	Business Meaning
Customer → Notification	1 : N	A customer receives many notifications over time
Customer → Dispute	1 : N	A customer may raise multiple disputes
Transaction → Dispute	1 : 0..N	A transaction may have zero or more disputes raised against it
Dispute → Dispute_Note	1 : N	A dispute accumulates investigation notes over its lifecycle
Dispute → Refund	0..1 : 0..1	An upheld dispute results in at most one refund
Transaction → Refund	0..1 : 0..1	A transaction may have at most one associated refund
User_Role → User_Role_Permission	1 : N	A role is granted permissions across one or more modules
User_Role → Internal_User	1 : N	A role is assigned to one or more internal users
Transaction / Dispute → Audit_Log	1 : N	Every state-changing event on these entities produces an audit log entry (polymorphic reference via entity_type/entity_id)